

Introduction

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Econ 126: Computational Macroeconomics

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Why Computational Methods for Macroeconomics

- Computational tools are indispensable for many areas of economic research and particularly for macroeconomics.
- Macroeconomists use computational tools to:
 - ① Solve and simulate complex dynamic models (often impossible with paper and pencil)
 - ② Manage and analyze data
- Undergraduate economic curriculum often shields students from this knowledge for a variety of reasons (e.g., pedagogical philosophy, perceived lack of math proficiency of students, cost to faculty of course design)

Example: RBC Model

- A three-equation real business cycle (RBC) model:

$$C_t^{-1} = \beta E_t [C_{t+1}^{-1} (\alpha A_{t+1} K_{t+1}^{\alpha-1} + 1 - \delta)] \quad (1)$$

$$C_t + K_{t+1} = A_t K_t^\alpha + (1 - \delta) K_t \quad (2)$$

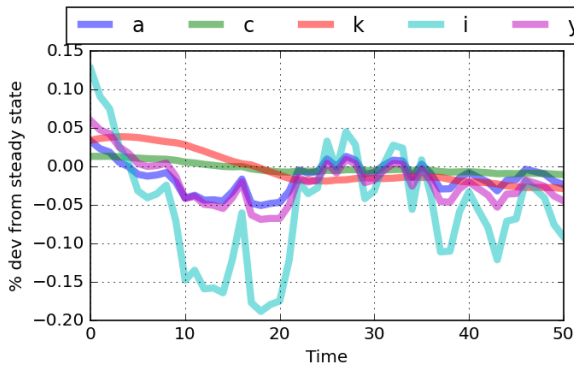
$$\log A_{t+1} = \rho \log A_t + \epsilon_{t+1} \quad (3)$$

- Equation (1) implies that C_t depends on the *expectation* of C_{t+1} .
- But Equation (1) also implies that the *expectation* of C_{t+1} depends on the *expectation* of C_{t+2} and so on.

Example: RBC Model

- So computing C_t requires also computing the expected path of $C_{t+1}, C_{t+2}, C_{t+3}, \dots$ given A_t and K_t .
- Solving the problem *analytically* (i.e., exactly with paper and pencil) would take pages of calculations and is error prone.
- Programmed properly, a computer can solve the problem *numerically* (i.e., approximately) in less than a second.

Figure 1: Simulated RBC model.



Why Python

- Macroeconomists have a lot of options:
 - ① Lower level, compiled languages: Fortran, C/C++
 - ② Higher level, scripting languages: Python, Matlab, Octave, Julia, R, Stata
- The compiled languages execute more quickly, but the time investment to write code is greater.

Why Python

- Among the scripting languages, Python has a lot of advantages:
 - ① Free and open source
 - ② Broad and active user base
 - ③ Many high quality libraries for numerical and statistical computing
 - ④ Versatile: Numerical applications, web scraping, email management, web development
- Versatility means that even if you don't become a researching macroeconomist or a financial engineer, you can still use Python.

- I use regularly use Python for purposes other than research:
 - Updating graphs in lecture slides automatically
 - Making animated videos for teaching purposes (e.g., <https://youtu.be/SdAuHSKtpmk>)
 - Sending personalized emails to students
 - Completing the DMCA takedown request form on Course Hero website
 - Designing a mural for a wall in my house.

Figure 2: Mural: Design concept. Colors are taken from the Sherwin Williams color catalog.

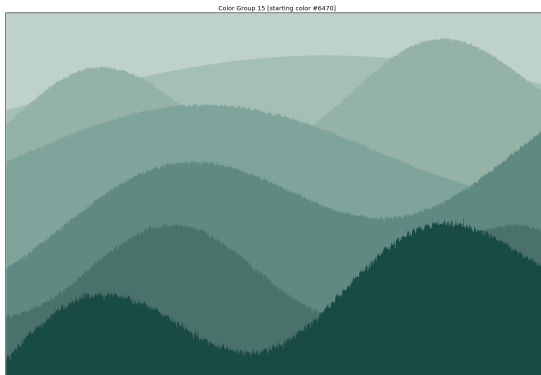


Figure 3: Mural: key coordinates.

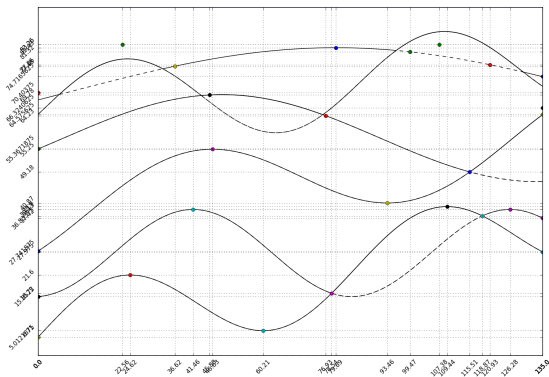


Figure 4: Mural: final result.



Overview of the Course

- In this course you will learn:
 - Basic Python programming skills
 - How to use Python to manage data and do basic statistics including linear regression
 - How to use Python to simulate linear dynamic models.
 - How to approximate nonlinear dynamic stochastic models.
 - The basics of the real business cycle (RBC) and new Keynesian modeling approaches
 - Critiques and criticisms of both business cycle modeling approaches

Overview of the Course

- The course presumes no programming experience
- My philosophy is that coding is like cooking: it's often sufficient to learn just what you need to know in order to make what you're trying to make